C1 - FIR-QP Pitch-Recovery Detailed Flow

Offline identification

FDDP nominal + C0-aligned pitch reference
$q_{ref}(t), \quad \theta_{ref}(t)$
fixed-phase C1 reference

Identify the empirical pulse response
α *pulse → root-pitch deviation*
$h = [h_0,\ldots,h_{M-1}]$
obtain FIR response coefficients

Fit and stabilise the AR disturbance model
$d_k = \Sigma a_j d_{k-j}$
check / contract the spectral radius

Online receding-horizon control

Measure pitch and form the reference-aligned error
$e_k = \theta_{meas,k} - \theta_{ref}(t)$

Subtract the known delayed response of past commands
$d_k = e_k - \Sigma h_i \alpha_{k-i}$

Forecast the free response and build the FIR dynamic matrix
$E_{free} = E_{AR} + E_{past}$
$E = E_{free} + SA$

Solve the tracking / magnitude / slew QP
$A^* = arg\ min\ q\|E_{free}+SA\|^2 + r\|A\|^2 + s\|DA\text{-}d_{prev}\|^2$
$|A_i| \le \alpha_{max}, \quad |\Delta\alpha| \le \Delta\alpha_{max}$

Apply only the first element of the optimal sequence
$\alpha_k = A_0^*$

Map to joint correction and position command
$\Delta q_k = G\alpha_k$
$q_{cmd} = q_{ref}(t) + \Delta q_k$

Compute and check acceptance metrics
IAE / peak and final pitch error / minimum contacts
$max\ |\Delta q|$
/ median and maximum QP solve time

Evidence boundary: C1 improves post-disturbance recovery relative to C0; it does not claim elimination of the initial peak at disturbance injection.